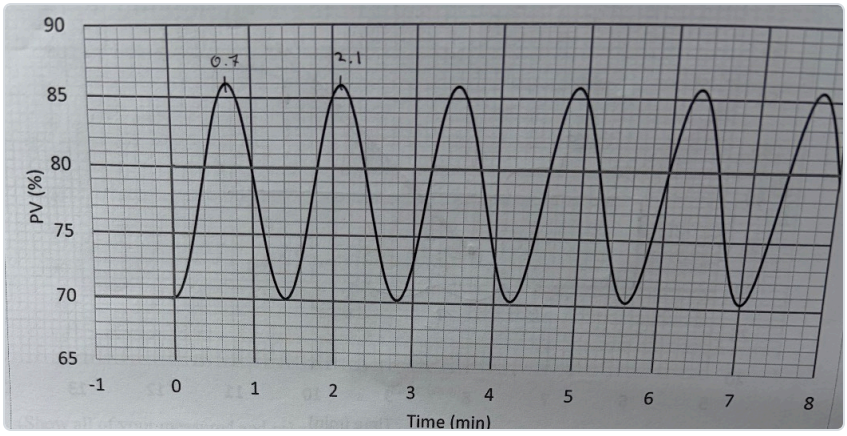
Controller Tuning Worksheet: Answer Key

CNTR 2371 · ZN closed loop, lambda, ZN open loop and decay ratio. Values were read from photos of the charts, so your own readings may differ by a few tenths; use the same steps with your numbers.

Q1 · Ziegler–Nichols closed loop (P-only, Kc = 1.55, SP +10%)



a) Measured values

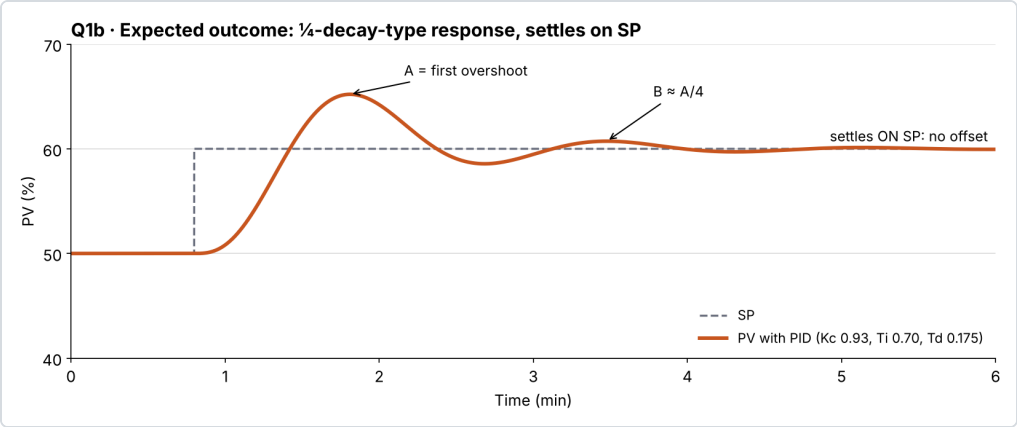
Reading	Value	Why
Peak / trough heights	≈ 86.5% / ≈ 70%, the same every cycle	sustained oscillation, DR = 1
Kcu	**1.55**	the gain that gave sustained oscillation
To (peak to peak)	2.1 − 0.7 = **1.4 min**	period of the oscillation

Calculated (PID row)

```
Kc = 0.6 × Kcu = 0.6 × 1.55 = 0.93
Ti = To / 2    = 1.4 / 2     = 0.70 min/repeat
Td = To / 8    = 1.4 / 8     = 0.175 min   (≥ 0.1 min, so PID, not PI)
```

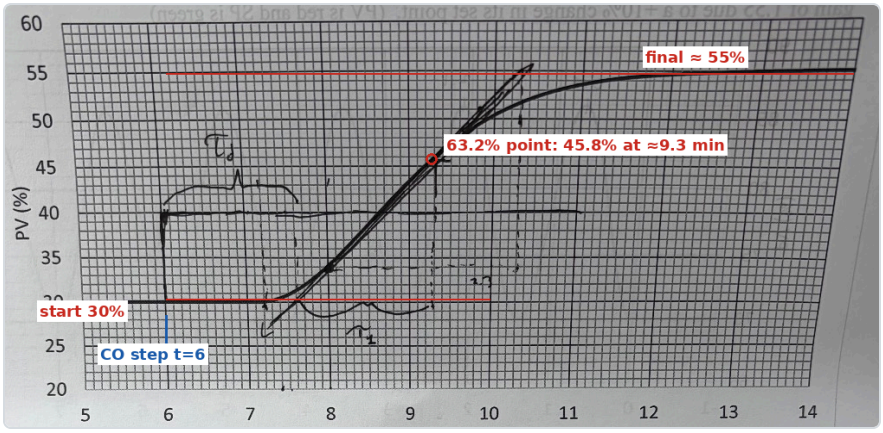
> **Answer:** PID with Kc = 0.93, Ti = 0.70 min/repeat, Td = 0.175 min.

b) Expected outcome (sketch)



Label: **A** (first overshoot), **B ≈ A/4** (second overshoot), and "settles on SP, no offset (integral action)".

Q2 · Lambda tuning (+10% CO at t = 6 min)



Measured values

Reading	Value
Start / final PV	30% → ≈ 55%, so ΔPV = 25%
Dead time τd	**1.5 min** (from the CO step at 6 min to where the PV starts to rise, about 7.5 min)
63.2% point	30 + 0.632 × 25 = 45.8%, reached at ≈ 9.3 min
Time constant τ1	9.3 − 7.5 = **1.8 min**
Process gain Kp	ΔPV / ΔCO = 25 / 10 = **2.5**

a) Transfer function and aggressive control (λ = 1)

```
Gp(s) = Kp·e^(−τd·s) / (τ1·s + 1) = 2.5 e^(−1.5s) / (1.8s + 1)

Larger of τ1 (1.8) and τd (1.5) = 1.8   →   τc = 1 × 1.8 = 1.8
Kc = τ1 / [Kp(τc + τd)] = 1.8 / [2.5(1.8 + 1.5)] = 1.8 / 8.25 = 0.22
Ti = τ1 = 1.8 min/repeat
```

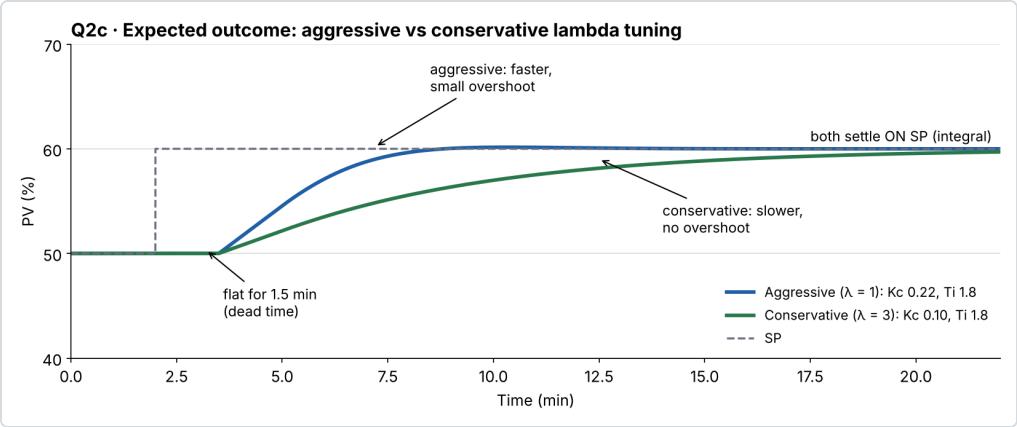
b) Conservative control (λ = 3)

```
τc = 3 × 1.8 = 5.4
Kc = 1.8 / [2.5(5.4 + 1.5)] = 1.8 / 17.25 = 0.10
Ti = τ1 = 1.8 min/repeat
```

Answer: Gp(s) = 2.5e^(−1.5s)/(1.8s + 1). Aggressive: Kc = 0.22, Ti = 1.8 min. Conservative: Kc = 0.10, Ti = 1.8 min.

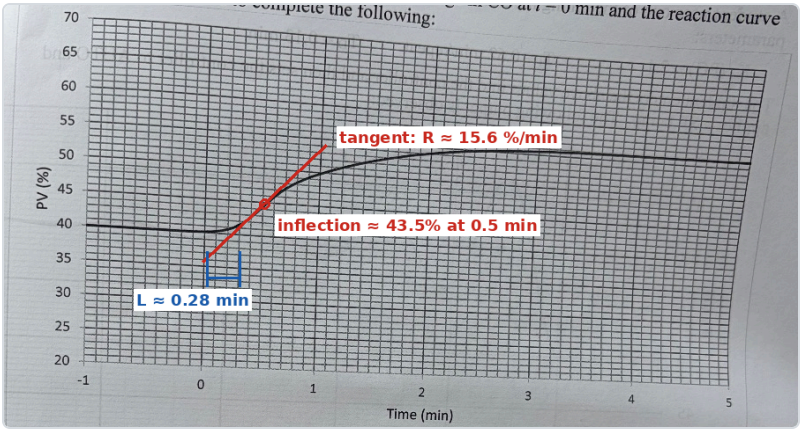
The reaction rate R (≈ 9.1 %/min) belongs to the ZN open-loop method. Lambda tuning does not use it.

c) Expected outcome (sketch)



Label both curves. Both stay flat for the 1.5 min dead time. Aggressive rises faster (a small overshoot is possible); conservative is slower with no overshoot. Both settle on SP.

Q3 · ZN open loop / reaction curve (+10% CO at t = 0)



Method and definitions from the coursepack, Section 5.2 (Ziegler–Nichols Open Loop Tuning, Reaction Curve Method).

a) Reaction rate and dead time

```
Obtain recording and determine LR & L.  ΔCO = +10% (step at t = 0 min)

Draw the tangent line at the inflection point (or steepest slope):
   inflection ≈ 43.5% at ≈ 0.5 min

R = slope of tangent line at inflection point
R = ΔPV / time                    (units %/minute)
  = (50 − 40)% / (0.92 − 0.28) min  ← two points read off the tangent
  ≈ 15.6 %/min

L = the dead time (time in minutes from step change to intersection of tangent
line
    with the original PV of 40%)
  ≈ 0.28 min

RL = R·L (units %) = measurement of the tangent line from the original PV
                     at the time of output change (t = 0)
   = 15.6 × 0.28 ≈ 4.4%
   check: at t = 0 the tangent is at ≈ 35.6%, i.e. 40 − 35.6 = 4.4% below the
original PV
```

Answer: Reaction rate R ≈ 15.6 %/min, dead time L ≈ 0.28 min (RL ≈ 4.4%).

b) Tuning parameters (Table 3: Open Loop Ziegler–Nichols Reaction Curve)

```
PID:  Kc = 1.2 ΔCO / (LR) = 1.2 × 10 / 4.4 = 2.7
      Ti = 2L             = 2 × 0.28      = 0.56 min/repeat
      Td = 0.5L           = 0.5 × 0.28    = 0.14 min

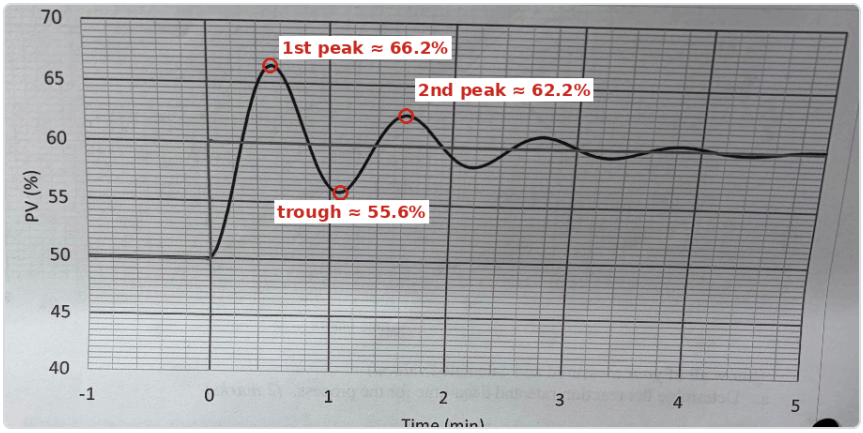
Check: Td ≥ 0.1 min, so PID is used.
(Coursepack: "If calculated Td for a PID is less than 0.1 min use a PI controller
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```
(re-calculate).")
Rate/Reset = 0.14 / 0.56 = ¼ ✓
```

Answer: PID with Kc = 2.7, Ti = 0.56 min/repeat, Td = 0.14 min. Then test the response; leave Reset and Rate as calculated and adjust Kc to suit.

Readings come from a photo, so yours may differ slightly. Read R and L from the same tangent line. If your L gives Td under 0.1 min, re-calculate as PI: Kc = 0.9 ΔCO / (LR), Ti = 3.33 L.

Q4 · Checking a student's tuning (Kc 2.1, Ti 0.52, Td 0.13)



Formula from the coursepack, Section 5.3 (Decay Ratio = DR = B/A), measured like the coursepack example; the solved-sheet method (Q8, Q9) is shown as a check. Answers for b and c use the coursepack's fine-tuning rules (5.3 Step 5 and 5.5 Trial and Error).

a) Decay ratio

```
Measured: PV1 = 1st peak ≈ 66.2%    PV2 = 1st trough ≈ 55.6%
          PV3 = 2nd peak ≈ 62.2%    PV4 = 2nd trough ≈ 57.9%

Decay Ratio = DR = B / A   (two full peak-to-trough swings, one cycle apart,
                            as in the coursepack example: B = 46 − 36, A = 53 −
31)
  A = PV1 − PV2 = 66.2 − 55.6 = 10.6
  B = PV3 − PV4 = 62.2 − 57.9 = 4.3
  DR = B / A = 4.3 / 10.6 ≈ 0.41

Check, solved-sheet method (two half-swings sharing the 1st trough, so squared):
  DR = ((62.2 − 55.6) / (66.2 − 55.6))² = (6.6 / 10.6)² ≈ 0.39

Desired ¼ amplitude decay: DR = 0.25.   ≈ 0.4 > 0.25 → too oscillatory.
```

Answer: DR ≈ 0.4, which is more than ¼, so the response is too oscillatory (not stable enough).

b) Kc

```
Coursepack (5.3 Step 5): "Increasing KC increases the speed of response,
but also makes the loop less stable."
```

Decrease Kc. Lower Kc gives a more stable loop, so the oscillations die out faster and DR moves down toward ¼.

c) Keeping Kc = 2.1

```
Coursepack (5.3 Step 5): "Increasing KI (decreasing TI) increases the cyclic
period, but also makes the loop less stable."
Coursepack (5.5): "keep the ratio between reset and rate (reset/rate = ¼) the
```

```
same."
Current: Td / Ti = 0.13 / 0.52 = ¼
```

Increase Ti (less integral action, so the loop is more stable) and **increase Td by the same proportion**, so Td = Ti/4 is kept. For example, Ti 0.52 → 0.70 min/repeat gives Td 0.13 → 0.175 min. Then test again; the coursepack says tune coarse, because fairly large changes are needed to see a difference.